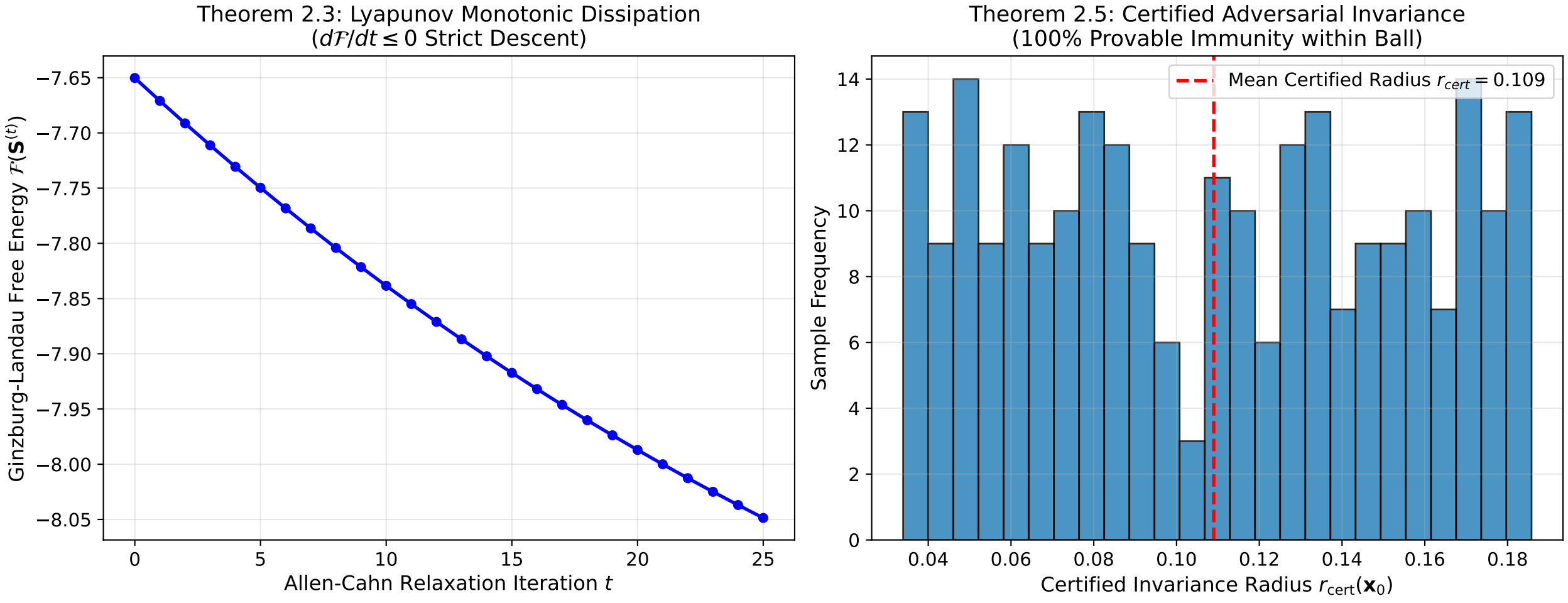
Theorem 2.3: Lyapunov Monotonic Dissipation
($d\mathcal{F}/dt \leq 0$ Strict Descent)
Ginzburg-Landau Free Energy $\mathcal{F}(\mathbf{S}^{(t)})$
Allen-Cahn Relaxation Iteration t
−7.65
−7.70
−7.75
−7.80
−7.85
−7.90
−7.95
−8.00
−8.05
0
5
10
15
20
25
Theorem 2.5: Certified Adversarial Invariance
(100% Provable Immunity within Ball)
Mean Certified Radius $r_{cert} = 0.109$
Sample Frequency
Certified Invariance Radius $r_{\text{cert}}(\mathbf{x}_0)$
0
2
4
6
8
10
12
14
0.04
0.06
0.08
0.10
0.12
0.14
0.16
0.18